MIL-F-1928E
4 April 1977
SUPERSEDING
MIL-F-1928D
2 October 1969

MILITARY SPECIFICATION

FILM LEADER, PHOTOGRAPHIC, SAFETY BASE

This Specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a film leader material that is to be used on film in conjunction with processing machines and for film identification marking.

1.2 Classification.

1.2.1 Types, subtypes, and classes. Film leaders covered by this specification shall be of the following types, subtypes, and classes. These groupings are based upon film size, perforations, and utilization:

TYPES

II	For 35mm film
III	For 16mm film
IV	For 70mm film
V	For 12.7 centimeters (cm)(5 inch) film
VI	For 24.13cm (9½ inch) film
VII	For 13.97cm (5½ inch) film
VIII	For 30.48cm (12 inch) film

SUBTYPES

A	Unperforated
B	Positive perforations
C	Negative perforations
D	Negative or positive perforations, one side only
E	Oversized positive perforations

FSC 6750

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CLASSES

1. Film leader, machine

2. Film leader, marking

3. Film leader, machine (colored)

4. Film leader, special purpose

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-621 Boxes, Wood, Nailed and Lock-Corner
PPP-B-636 Boxes, Shipping, Fiberboard
PPP-B-1055 Barrier Material, Waterproofed, Flexible

Military

MIL-L-10547 Liners, Case, and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible

STANDARDS

Military

MIL-STD-105 Sampling Procedures and Tables For Inspection by Attributes
MIL-STD-129 Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE STANDARDS

PH1.20 70mm Unperforated and Perforated Film for Camera Other than Motion-Picture Cameras Dimensions For
PH1.25 Safety Photographic Film, Specifications For
PH1.28 Photographic Film for Archival Records, Silver Gelatin Type, on Cellulose Ester Base Specifications For
PH1.31 Brittleness of Photographic Film, Method For Determining The

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PH22.37	Raw Stock Cores for 35mm Motion Picture Film, Dimensions of
PH22.38	Raw Stock Cores for 16mm Motion Picture Film, Dimensions of
PH22.93	35mm Motion Picture Film, Perforated BH, Dimensions for
PH22.109	16mm Motion Picture Film, Perforated 1R, Dimensions for
PH22.110	16mm Motion Picture Film, Perforated 2R, Dimensions for
PH22.139	35mm Motion Picture Film, Perforated KS, Dimensions for

(Copies of ANSI Standards may be obtained from the American National Standards Institute, 1430 Broadway, New York, New York 10018)

3. REQUIREMENTS

3.1 Materials. When materials, which are not specifically designated, are used, they shall be suitable for the purpose.

3.1.1 Base material. The leader shall be of safety-base material conforming to ANSI PH1.25 without emulsion coating.

3.2 Perforations. Unless otherwise specified by the procuring activity, leader shall be supplied as Subtype A. Perforations, when specified, shall be in accordance with the standards referenced in Table I.

3.3 Leader widths. Leader widths shall be cut in accordance with Table I.

3.4 Length. Length will be as specified in the individual contract. Tolerance is +3.048 Meters (+10 feet) and -0 Meters (M) (-0 feet) (see 6.2).

*3.5 Cores or spools. Leader shall be wound on either cores or spools. Leader wound on cores shall be in accordance with Table I. When spools are specified by the procuring activity, the nominal size of the spool employed shall correspond to the designated leader length.

*3.6 Flexibility. The leader shall be tested to determine flexibility and meet the requirements as specified in ANSI Standard PH1.28.

3.7 Brittleness. The leader shall be tested in accordance with Standard PH1.31, Method B. The minimum average wedge separation for acceptable leader brittleness shall be as follows:

Total thickness	Under 0.13mm (0.005 in)	0.13-0.18mm (0.005-0.007 in)	Over 0.18mm (0.007 in)
Minimum average wedge separation	2.54 milimeter (0.10 in)	3.81mm (0.15 in)	5.08mm (0.20 in)

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3.8 Chemical resistance. When Tested in accordance with 4.3.2.1 and after the leader has cooled to room temperature, any evidence of curl, warp, buckle, crack, expansion, shrinkage, greater than one percent, or other damage which will materially affect the serviceability or make the material unfit for use, shall be cause for rejection.

3.9 Classes.

*3.9.1 Class 1. Class 1 leader shall be opaque with each side a different color to be easily differentiated. The thickness shall be 0.1016mm (0.0040 in) ± 0.007mm (0.0003 in) or 0.2032mm (0.0080 in) ± 0.0127mm (0.0005 in). If other than those thickness specified are required, the procuring activity will so indicate.

*3.9.2 Class 2. Class 2 leader shall be optically homogenous, transparent, free from scratches and abrasion marks, noncolor selective, uncoated, and have a maximum transmission density of 0.10. The thickness shall be 0.1016mm (0.0040 in) ± 0.0076mm (0.0003 in), or 0.1270mm (0.0050 in) ± 0.0127mm (0.0005 in), or 0.1778mm (0.0076 in) ± 0.0127mm (0.0005 in) as specified by the procuring activity.

*3.9.3 Class 3. Class 3 film leader shall be transparent and uncoated. It shall be colored, as specified by the procuring activity, to aid in its identification and for ease of observation. The thickness shall be 0.1778mm (0.0070 in) ± 0.0127mm (0.0005 in), or 0.2413mm (0.0095 in) ± 0.0127mm (0.0005 in) as specified by the procuring activity.

*3.9.4 Class 4. Class 4 leader shall be completely transparent without tint, or shall be opaque, as specified by the procuring activity. The thickness shall be as specified in the contract or purchase order (see 6.2).

3.10 Workmanship. Workmanship shall be in accordance with high-grade commercial practice covering the manufacture of motion picture film base.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Sampling for initial approval. When samples are required by the procuring activity in order to establish a particular film leader as suitable for classification under any one of the types and classes listed herein, the following procedure shall be followed. The manufacture shall supply a sufficient quantity of film leader to accomplish all tests required to determine compliance with the requirements set forth herein. Such samples

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TABLE I DATA FOR FILM LEADER

Type	Nominal Size	Width Requirement	Perforations		Core
			Subtype	Requirement	
II	35mm	ANSI PH22.139	A	Not Applicable	ANSI PH22.37
			B	ANSI 22.139 ANSI 22.93	
			C	ANSI PH22.93	
			E	No ANSI Standard available	
III	16mm	ANSI PH22.110	A	Not Applicable	ANSI PH22.38
			B	ANSI PH22.110	
			C	ANSI PH22.110	
			D	ANSI PH22.109	
			E	No ANSI Standard available	
IV	70mm	ANSI PH1.20	A	Not Applicable	None
			B	ANSI PH1.20	
			C	ANSI PH1.20	
V	5 in	4.960±0.010 in	A	Not Applicable	None
VI	9½ in	9.460±0.010 in	A	Not Applicable	None
VII	5½ in	5.507±0.010 in	A	Not Applicable	2 3/4 in ID Paperboard
VIII	12 in	12.00 ± 0.010 in	A	Not Applicable	2 3/4 in ID Paperboard

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shall be clearly marked in the nomenclature of this specification, with the type, subtype and class under which the manufacturer intends to submit the film leader.

4.2 Sampling for inspection and acceptance. Unless otherwise specified, sampling for inspection and acceptance shall be performed in accordance with the provisions set in MIL-STD-105. The manufacturer's sampling and inspection procedures may be utilized providing such sampling and inspection methods will assure quality equal to or better than that obtained by the following provisions set forth in MIL-STD-105. For purposes of sampling, an inspection lot shall consist of all material of the same type, subtype, and class submitted for inspection and delivery at one time.

4.2.1 Inspection of materials and components. In accordance with 4.1 above, the supplier is responsible for insuring that materials and components used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or, if none, in accordance with this specification. In the event of conflict, this specification shall govern.

4.2.2 Inspection and quality of the end item.

4.2.2.1 Quality of end item. The quality of the end item is the manufacturer's responsibility and can be determined by any means of inspection and testing at the option of the manufacturer. However, the quality of the end item must be as good or better than the quality obtained by the inspection and tests set forth in the following paragraphs.

4.2.2.2 Examination of the end item. Examination of the end item shall be made in accordance with the classification of defects, inspection levels and acceptable quality levels (AQLs) set forth below. Not more than 5 rolls of leader shall be drawn from any one shipping container from each lot, type, subtype, and class of leader offered for acceptance for visual and dimensional characteristics. The lot size for purposes of determining sample size in accordance with MIL-STD-105 shall be expressed in units of rolls for examination under 4.2.2.2.1, 4.2.2.2.2, and 4.2.2.2.3, and in units of shipping containers for examination in 4.2.2.2.4.

4.2.2.2.1 Examination of the end item for defects in appearance and workmanship. The sample unit for this examination shall be 30.48 linear centimeters (CM) (1 linear foot) of leader. Not more than five sample units shall be drawn from any one package or roll.

Examine	Defect
Appearance and workmanship	Leader unevenly and not completely colored (when required).
	Surface shows scuffs, scratches, foreign matter or abrasions.
Perforations (when required)	Holes not cleanly punched.

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4.2.2.2.2 Examination of the end item for dimensional defects. The sample unit for this examination shall be 30.48 linear (CM) (1 linear foot) of leader.

Examine	Defect
Width and perforations	Not within tolerances of Table I.

4.2.2.2.3 Examination of the end item for defects in construction and length. The sample unit for this examination shall be one roll of leader.

Examine	Defect
Leader	Rolls not evenly or neatly wound; not wound on specified core, dimensions of core not as specified, core cracked, broken, chipped, mutilated or collapsed. Not spooled as specified. Not perforated as specified. Roll not continuous.
Length of roll	Not in conformance with specified contract length and allowable deviation.

4.2.2.2.4 Examination of preparation for delivery. An examination shall be made to determine that marking, materials, workmanship, contents, and weights comply with the requirements of this specification. Defects shall be scored in accordance with the classification below. The sample unit shall be one shipping container, fully packed, selected just prior to the closing operation. Shipping containers fully prepared for delivery shall be examined for closure defects.

Examine	Defect
Packaging and preservation	Not as specified, not in accordance with contract requirements. Unit packages not uniform. Containers of improper construction.
Packing	Not packed as specified. Packing material not as specified; closures not accomplished by specified or required methods or materials.
	Number of rolls per carton less than indicated or specified.
Markings (Interior and exterior)	Interior labels not correct color. Date on labels not in accordance with contract specifications. Labels not properly attached.

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	Exterior labels or markings illegible, incorrect, incomplete, omitted, or not in accordance with contract requirements.
Weight (Interior and exterior)	Gross or net weight exceeds contract requirements.

4.2.2.2.5 Inspection levels and acceptable quality levels (AQLs) for examinations. The inspection levels, for determining the sample size and the acceptable quality levels (AQLs) expressed in defects per 100 units, shall be as follows:

Examination	Inspection Level	AQL
4.2.2.2.1	I	1.0
4.2.2.2.2	S-3	2.5
4.2.2.2.3	S-2	2.5
4.2.2.2.4	S-2	4.0

4.2.3 Testing of the end item. Testing of the end item shall be conducted in accordance with Table II for the characteristics as indicated therein on each lot submitted for acceptance. The sample unit, as referenced herein, is defined as the amount of material required to perform all tests one time. The sample unit for all testing shall consist of 3 linear feet of leader. Five sample units, each drawn from a different shipping container, shall be tested with no evidence of failure to meet sample unit or average requirements, as applicable.

4.3 Acceptance tests. Acceptance tests shall consist of functional and periodic sampling tests.

4.3.1 Functional tests. At the request of the procuring activity, a 304.8m (1,000 foot) length of leader, representative of each type and class on order, shall be furnished the procuring activity for tests in film processing machines to determine compliance with the requirements of this specification.

4.3.2 Periodic sampling tests. Periodic sampling tests shall consist of the following:

4.3.2.1 Chemical resistance test. Not less than 3.04m (ten feet) of each type of leader on order shall be soaked in a sodium hydroxide solution with a pH between 11.5 and 12.0 at 51° celsius (C) (125° Fahrenheit [F]) for one hour after which it shall be dried for two hours in a dry incubator at a temperature of 71.1°c (160°F).

4.3.2.2 Flexibility test. The leader shall be tested to determine flexibility in accordance with ANSI PH1.28.

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4.3.2.3 Brittleness test. The leader shall be tested to determine brittleness in accordance with ANSI PH1.31, Method B.

4.3.3 Rejection and retest. Where one item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure are determined and corrected.

5. PACKAGING.

*5.1 Preservation. Each roll of film leader shall be preserved in a metal or rigid plastic container of suitable size and design manufactured to the best commercial standards. The materials used shall protect the product against deterioration throughout its period of expected military storage.

5.2 Packaging.

5.2.1 Level A. Leader shall be preserved in accordance with 5.1 and packaged in a unit corrugated or simple paperboard carton in accordance with the manufacturer's best commercial practice for the product. All materials used in preservation and packaging of film shall be relatively photographically inert and shall be of the proper strength and design with adequate pads or blocking so that when packed either Level "A", Level "B" or Level "C" shall insure adequate protection of the product.

5.3 Packing. Packing shall be Level A, B or C as specified (see 6.2).

*5.3.1 Level A. Film packaged in accordance with 5.2 shall be packed in exterior-type shipping containers in accordance with PPP-B-636, Type CF, variety SW, class weather-resistant, grade V3c. Alternatively, containers conforming to PPP-B-621, class 2, Style IV may be used. Containers conforming to PPP-B-621 shall be provided with a case liner conforming to PPP-B-1055 and fabrication in accordance with MIL-L-10547. As for as practical, exterior containers shall be of uniform shape and size, be of minimum cube and tare consistant with the protection required, and contain identical quantities. The gross weight of each pack shall be limited to 34 Kilograms (K)(75 pounds) for the container specified in PPP-B-636 and 90.7K (200 pounds) for the container specified in PPP-B-621. Strapping and closure shall be in accordance with the appendix of the applicable container specification.

*5.3.2 Level B. Film packaged in accordance with 5.2.1 shall be packed in exterior-type shipping containers in accordance with PPP-B-636, Type CF, variety SW, class domestic, grade 350, special requirements use. Alternatively, container conforming to PPP-B-621, Class I, Style IV may be used.

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TABLE II - INSTRUCTIONS FOR TESTING (SAMPLE UNIT)

Characteristic	Spec Reference		Rqmts Appl to		Number Determinations Per Sample Unit	Results Report As	
	Rqmts	Test Method	Indiv Unit	Lot Average		Pass or Fail	Numerically to Nearest
Flexibility	3.6	4.3.2.2	X	-	1	X	
Brittleness	3.7	4.3.2.3	X	-	1	X	
Chemical Resistance	3.8	4.3.2.1	X	-	1	X	
Thickness							
Class 1	3.9.1	3.9.1	X	-	Avg of 3	-	0.0001 inch
Class 2	3.9.2	3.9.2	X	-	Avg of 3	-	0.0001 inch
Class 3	3.9.3	3.9.3	X	-	Avg of 3	-	0.0001 inch
Class 4	3.9.4	3.9.4	X	-	Avg of 3	-	0.0001 inch
1/Functional Test	4.3.1	4.3.1	X	-	1	X	

1/Contractor shall furnish a certificate of compliance that material complies with this requirement.

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As far as practical, exterior containers shall be of uniform shape and size. be of minimum cube and tare consistent with the protection required, and contain identical quantities. The gross weight of each pack shall be limited to 90 pounds. Container strength and design shall be adequate to insure protection against damage during multiple domestic shipments, handling, and covered storage.

5.3.3 Level C. Film packaged in accordance with 5.2, requiring overpacking for acceptance by the carrier, shall be packed in exterior-type shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery. Container shall meet Uniform Classification rules or other regulations of common carriers as applicable to the mode of transportation.

*5.4 Labeling. A label, in accordance with the label layout shown in figure 1, shall be firmly attached to each can and unit package. Labels shall be printed in heavy black on white paper stock and shall be of an appropriate size to permit easy reading of the print. Commercial labels containing the same information with modified format and/or additional information will be acceptable.

5.5 Marking

5.5.1. Exterior shipping container. Each exterior container shall be marked in accordance with MIL-STD-129, except that the dimensions of the container and preservation method need not be included. Abbreviated nomenclature may be used when necessary. The complete lot, batch or control number, and size shall be marked on the shipping container, unless otherwise specified by the procuring activity. The nomenclature shall be:

PHOTOGRAPHIC FILM MATERIAL
STORE IN COOL DRY LOCATION

COMMERCIAL NAME
OF PRODUCT

TYPE SUBTYPE CLASS
LENGTH
WIDTH
PERFORATIONS

TYPE OF WINDING OR SPOOL

STOCK NUMBER

FILM LEADER, PHOTOGRAPHIC, SAFETY BASE
SPEC NUMBER

MANUFACTURER'S NAME AND CLAUSES, ETC

FIGURE 1 - LABEL LAYOUT

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6 NOTES AND CONCLUDING MATERIAL.

6.1 Intended use. Leaders covered by this specification are intended for use as:

a. Leaders or trailers when threading roll film through processing machine.

b. A means of marking rolls of film for identification.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Type, class, and color of leader required, subtype when perforations are desired.

c. Lengths of film leader required (see 3.4).

d. If functional test sample is required.

e. Level of packing required (see 5.3).

f. Core, spool, and length, if other than as specified in Table 1.

g. Thickness required (see 3.9).

6.3 Asterisks. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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